

DEQ SITE IDENTIFICATION FORM

Complete the following information and provide the attachments listed below for each new site or existing site that has not had a Site Identification Form with attachments submitted to the DEQ since January 1, 1998. Please amend any previous Site Identification Forms when it becomes known that any of the information requested below has changed (i.e. new land owner).

FACILITY NAME City of Cadillac WWTP		NPDES or COC PERMIT NUMBER MI0020257	
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SITE IDENTIFICATION FORM					
A. Site ID	Site ID # 21N09W25-AB01	Street or Route # West of Seeley Rd, North of 50 Rd			
B. County/Township:	County Wexford	Township Clam Lake			
C. State Planar Coordinates:	SE 1/4	SE 1/4	Section 25	Town T21N	Range R09W
D. Lat/Long (nearest second):	Latitude 44° 10' 53.93"N	Longitude 85° 20' 14.82"W		Method Used (i.e. GPS, topo) Google Earth	

E. Site Ownership	Name of Owner Alan Benson		Phone 231-920-6783
	Street or P.O. Box 11799 East 50 Mile Rd		
	City or Town Cadillac	State MI	Zip 49601
	Is there a written agreement from the owner to land apply? ☒ Yes ☐ No, written agreement is required		Previous landowners name Hartley Benson Property purchase date N/A

	Name of Lessee (if different)	Phone
	Street or P.O. Box	
	City or Town	State Zip

F. Site Characteristics	Site Type ☒ Agricultural ☐ * Reclamation Site ☐ Forest ☐ ** Lawn/Home Garden *Separate approval is required **Requires EQ Biosolids	Soils Information Date of last soils analysis <u>11-15-2019</u> *Phosphorous <u>76</u> units PPM Potassium <u>144</u> units PPM pH <u>7.0</u> Soil type(s) <u>Sandy</u> *300 lb/ac Bray P1 maximum								
	Expected crops/vegetation to be grown <table style="width: 100%; border-collapse: collapse;"> <tr> <td style="width: 30%;">Crop/Vegetation</td> <td>Nitrogen Requirement</td> </tr> <tr> <td><u>Corn</u></td> <td><u>135</u> lbs. N/acre</td> </tr> <tr> <td>_____</td> <td>_____ lbs. N/acre</td> </tr> <tr> <td>_____</td> <td>_____ lbs. N/acre</td> </tr> </table>	Crop/Vegetation	Nitrogen Requirement	<u>Corn</u>	<u>135</u> lbs. N/acre	_____	_____ lbs. N/acre	_____	_____ lbs. N/acre	Percent slope Highest percent slope of the site? ☐ 0-6% ☐ 6-12% ☒ *Higher (Requires an approved site reclamation plan)
Crop/Vegetation	Nitrogen Requirement									
<u>Corn</u>	<u>135</u> lbs. N/acre									
_____	_____ lbs. N/acre									
_____	_____ lbs. N/acre									
	Total acreage of site <u>46.2</u> Acreage used for crops <u>46.2</u>	If agricultural, is the site tilled? ☐ Yes ☒ No ☐ NA								
	Public Exposure Is this land with a high potential for public exposure? (See Definitions section on Page 19) ☐ Yes ☒ No									

On an attached sheet provide a copy of the following:

- A Farm Services Agency map (formerly ASCS) of the proposed application site with property boundaries, surface waters, and discharge location of field tiles when present.
- A copy of the soil test analysis that is no older than 2 years old at the time of land application.
- A copy of the plat map (or other map showing property dimensions) with the proposed land application site highlighted.
- A soil survey map with the land application site identified.



CITY OF CADILLAC WASTEWATER TREATMENT PLANT

BIOSOLIDS RECYCLING PROGRAM PARTICIPATION

I hereby agree to participate in the City of Cadillac WWTP biosolids recycling program. I understand that biosolids will be applied to my land in accordance with federal EPA regulations and guidelines of the Michigan Department of Environmental Quality.

I understand that biosolids have been analyzed to be sure they can be applied to agricultural land in accordance with federal regulations and MDEQ guidelines. I understand that biosolids contain some heavy metals and that they must be below the EPA allowable limits for land application. I understand that the nutrients and/or the metal concentration may limit the frequency of biosolids application to my land.

My land is used for the production of agricultural crops. Information concerning crop and site restrictions has been explained to me. I agree to allow representatives of the MDEQ, Cadillac Waste Water Treatment Plant and the City's Injection contractor to have access to my land to obtain soil samples and for application of biosolids.

If I decide to cancel this agreement, I will give 60 days advance notice to the City of Cadillac WWTP.

Landowner: Alan Benson Signed: Alan Benson

Address: 11299 East 50mi Rd Date: 10-29-19

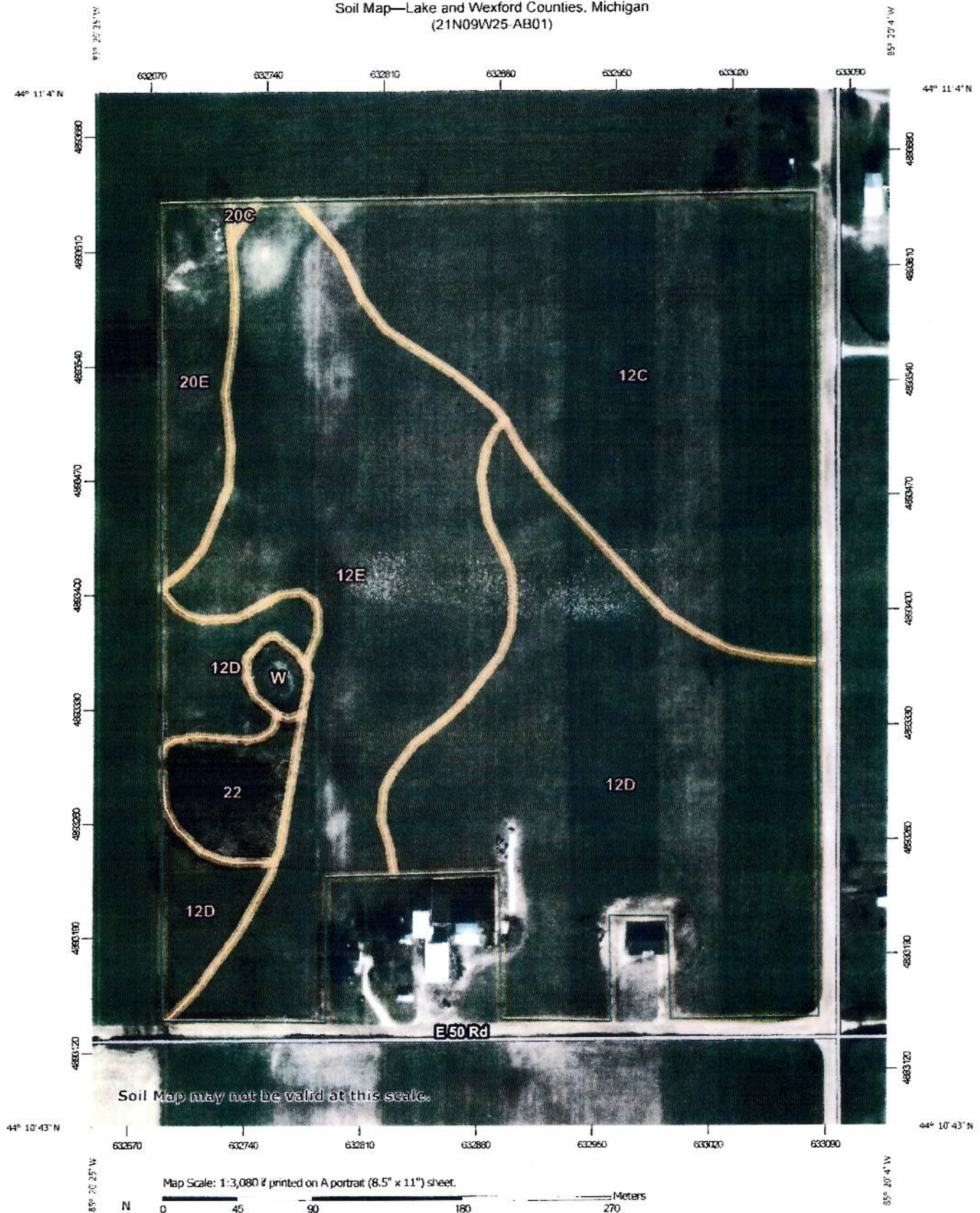
Township: Clam Lake Section: 25 Acres: 50

City WWTP Field Number: 21N09W25-AB01

If Rented, Farmer: _____

Address: _____

Soil Map—Lake and Wexford Counties, Michigan
(21N09W25-AB01)



Soil Map may not be valid at this scale.

Map Scale: 1:3,080 if printed on A portrait (8.5" x 11") sheet.

0 45 90 180 270 Meters

0 100 200 400 600 Feet

Map projection: Web Mercator Corner coordinates: WGS84 Edge ticks: UTM Zone 16N WGS84



Natural Resources
Conservation Service

Web Soil Survey
National Cooperative Soil Survey

10/28/2019
Page 1 of 3

MAP LEGEND

	Area of Interest (AOI)		Spot Area
	Area of Interest (AOI)		Stony Spot
	Soils		Very Stony Spot
	Soil Map Unit Polygons		Wet Spot
	Soil Map Unit Lines		Other
	Soil Map Unit Points		Special Line Features
	Special Point Features		Water Features
	Blowout		Streams and Canals
	Borrow Pit		Transportation
	Clay Spot		Rails
	Closed Depression		Interstate Highways
	Gravel Pit		US Routes
	Gravelly Spot		Major Roads
	Landfill		Local Roads
	Lava Flow		Background
	Marsh or swamp		Aerial Photography
	Mine or Quarry		
	Miscellaneous Water		
	Perennial Water		
	Rock Outcrop		
	Saline Spot		
	Sandy Spot		
	Severely Eroded Spot		
	Sinkhole		
	Slide or Slip		
	Sodic Spot		

MAP INFORMATION

The soil surveys that comprise your AOI were mapped at 1:15,800.

Warning: Soil Map may not be valid at this scale.

Enlargement of maps beyond the scale of mapping can cause misunderstanding of the detail of mapping and accuracy of soil line placement. The maps do not show the small areas of contrasting soils that could have been shown at a more detailed scale.

Please rely on the bar scale on each map sheet for map measurements.

Source of Map: Natural Resources Conservation Service
Web Soil Survey URL:
Coordinate System: Web Mercator (EPSG:3857)

Maps from the Web Soil Survey are based on the Web Mercator projection, which preserves direction and shape but distorts distance and area. A projection that preserves area, such as the Albers equal-area conic projection, should be used if more accurate calculations of distance or area are required.

This product is generated from the USDA-NRCS certified data as of the version date(s) listed below.

Soil Survey Area: Lake and Wexford Counties, Michigan
Survey Area Data: Version 13, Sep 16, 2019

Soil map units are labeled (as space allows) for map scales 1:50,000 or larger.

Date(s) aerial images were photographed: May 9, 2014—Sep 19, 2016

The orthophoto or other base map on which the soil lines were compiled and digitized probably differs from the background imagery displayed on these maps. As a result, some minor shifting of map unit boundaries may be evident.

Map Unit Legend

Map Unit Symbol	Map Unit Name	Acres in AOI	Percent of AOI
12C	Emmet-Montcalm complex, 6 to 12 percent slopes	13.4	29.0%
12D	Emmet-Montcalm complex, 12 to 18 percent slopes	16.2	35.0%
12E	Emmet-Montcalm complex, 18 to 40 percent slopes	12.7	27.5%
20C	Montcalm-Graycalm complex, 6 to 12 percent slopes	0.0	0.1%
20E	Montcalm-Graycalm complex, 12 to 30 percent slopes	2.2	4.7%
22	Tawas-Roscommon association	1.3	2.9%
W	Water	0.3	0.7%
Totals for Area of Interest		46.2	100.0%

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www.kettunencenter.org
(231) 829-3421



CLAM LAKE T21N • R9W

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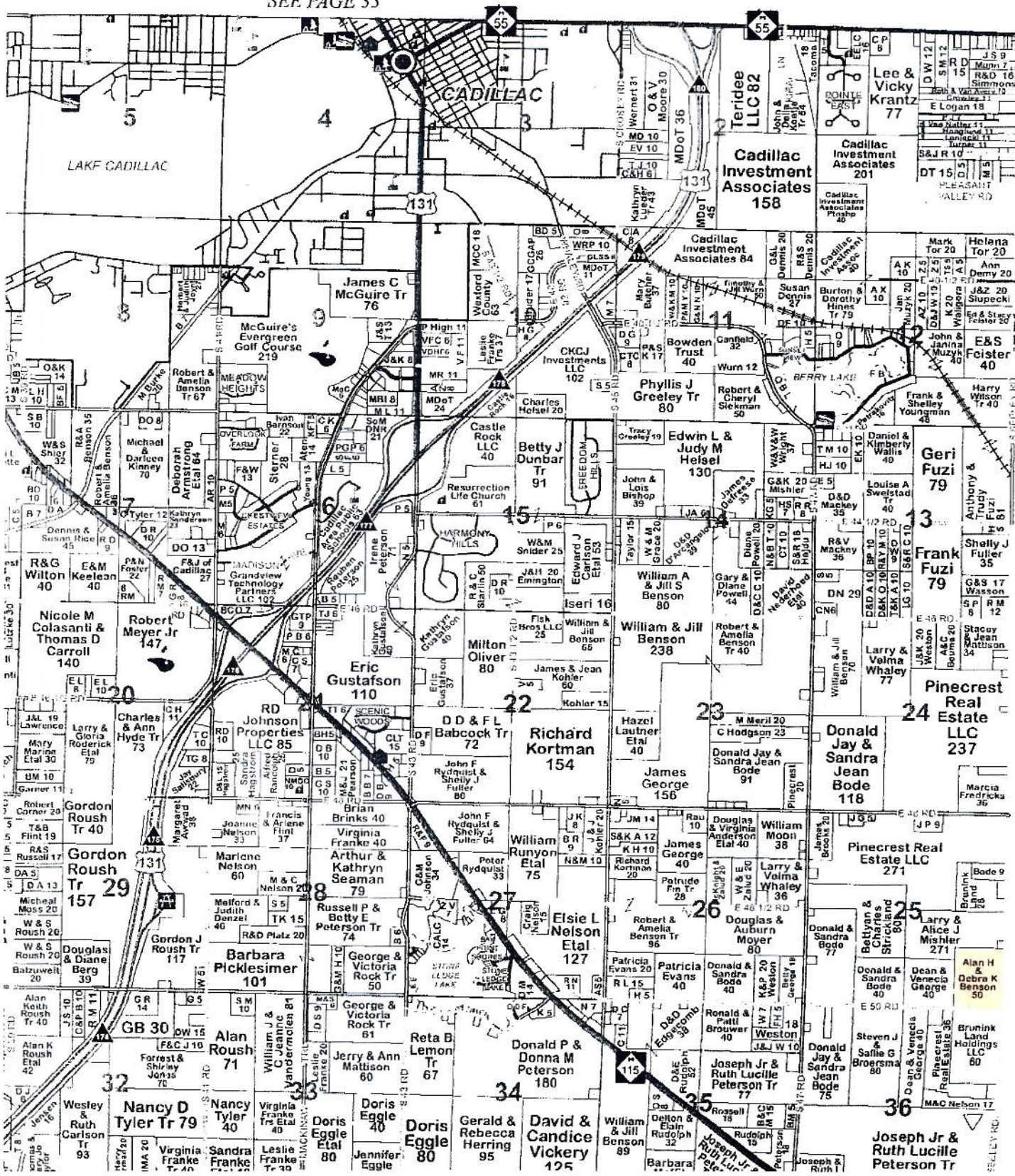
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MISSAUKIE COUNTY

Report Number
F23290-0742
Account Number
15170



3505 Conestoga Dr.
Fort Wayne, IN 46808
260.483.4759
algreatlakes.com

To: CITY OF CADILLAC
200 N LAKE ST
CADILLAC, MI 49601-2800

For: FALL 2023 APPLICATIONS

Attn: PATRICK MILLER

SOIL TEST REPORT

Date Received: 10/17/2023
Date Reported: 10/19/2023 Page: 1 of 1

Sample ID	Lab Number	Organic Matter %	Phosphorus		Potassium K ppm	Magnesium Mg ppm	Calcium Ca ppm	Sodium Na ppm	Soil pH	Buffer pH	CEC meq/100g	Percent Cation Saturation				
			Bray-1 Equiv ppm-P	Bray P2 ppm-P								% K	% Mg	% Ca	% H	% Na
AB01WEX	9825	2.1	110 VH	147 VH	172 H	165 H	850 M		7.0		6.1	7.3	22.7	70.1		
AB02MIS	9826	2.1	169 VH	232 VH	247 VH	195 H	950 M		7.0		7.0	9.0	23.2	67.8		
AB01MIS	9827	2.3	152 VH	222 VH	186 H	175 H	950 M		6.8		6.9	6.9	21.2	68.9	3.0	
DG01WEX	9828	2.4	57 VH	69 H	100 M	185 H	900 M		6.7	6.9	7.5	3.4	20.6	60.0	16.0	
DG01MIS	9829	2.3	82 VH	99 VH	141 H	195 H	950 M		6.9		6.8	5.3	23.8	69.5	1.5	

VL = Very Low L = Low M = Medium H = High VH = Very High

Sample ID	Sulfur S ppm	Zinc Zn ppm	Manganese Mn ppm	Iron Fe ppm	Copper Cu ppm	Boron B ppm	Soluble Salts (1:2) mmhos/cm	Nitrate NO ₃ -N ppm	Ammonium NH ₄ -N ppm	Bicarb-P P ppm				Comments

Report reviewed and approved by our professional agronomy staff.

AK-R

Report Number
F23290-0742
Account Number
15170



3505 Conestoga Dr.
Fort Wayne, IN 46808
260.483.4759
algreatlakes.com

To: CITY OF CADILLAC
200 N LAKE ST
CADILLAC, MI 49601-2800

For: FALL 2023 APPLICATIONS

Attn: PATRICK MILLER

SOIL FERTILITY RECOMMENDATIONS

Date Received: 10/17/2023
Date Reported: 10/19/2023

Page: 1 of 1

Sample ID	Intended Crop	Previous Crop	Yield Goal	Lime Tons/A	Nitrogen N lb/A	Phosphate P ₂ O ₅ lb/A	Potash K ₂ O lb/A	Magnesium Mg lb/A	Sulfur S lb/A	Zinc Zn lb/A	Manganese Mn lb/A	Iron Fe lb/A	Copper Cu lb/A	Boron B lb/A
AB01WEX	Corn	Corn	180 bu	0.0	235	0	45	0						
AB02MIS	Corn	Corn	180 bu	0.0	235	0	0	0						
AB01MIS	Corn	Corn	180 bu	0.0	230	0	45	0						
DG01WEX	Corn	Corn	180 bu	0.0	230	35	140	0						
DG01MIS	Corn	Corn	180 bu	0.0	230	0	55	0						